

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset5_expanded • Generated: 2026-04-14 20:57:29

I. CASE IDENTIFICATION	
Dataset ID	dataset5_expanded
Scan ID	ds-005-exp
Source File	windows_soc_data_5_expanded.csv
Primary Threat	Multi-Stage Persistent Intrusion with Credential Theft and Exfiltration
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	160
Aggregate Risk Score	7,400
Threat Level	HIGH

TABLE OF CONTENTS		
I	Executive Summary	3
II	Case Identification & Background	4
III	Description of Evidence & Chain of Custody	4
IV	Forensic Methodology & Tools	5
V	Statistical & Graphical Analysis	6
VI	Brute Force Attack - Detailed Analysis	8
VII	Log Tampering - Detailed Analysis	9
VIII	Lateral Movement - Detailed Analysis	10
IX	Privilege Escalation - Detailed Analysis	11
X	Malicious Service Installation - Analysis	12
XI	Host Compromise Matrix	13
XII	Attack Chain Reconstruction	14
XIII	Indicator of Compromise (IOC) Register	15
XIV	Threat Actor Profile & Attribution	16
XV	Risk Heat Map - Host x Technique Exposure	17
XVI	MITRE ATT&CK; Framework Coverage	18
XVII	Remediation Recommendations	19
XVIII	Expert Opinion & Conclusion	21
XIX	Annexure A - Forensic Environment Specification	22
XX	Annexure B - Extended Event Log Samples	23
XXI	Annexure C - Glossary of Terms	27

I

EXECUTIVE SUMMARY

- EXECUTIVE SUMMARY
- A multi-stage cyber attack campaign involving 8 coordinated attack chains was detected across multiple systems and users, indicating a persistent and organized intrusion.
- The attack exhibits Advanced Persistent Threat (APT) characteristics, with stealthy and low-and-slow techniques used to maintain long-term access and evade detection.
- Initial access was gained through exploitation of a public-facing application (IIS), followed by deployment of web shells and reverse shells for remote control.
- PowerShell encoded commands and fileless execution techniques were widely used to execute payloads and reduce forensic traces.
- Persistence was established using registry run keys and scheduled tasks, ensuring continued access even after system restarts.
- Credential access included LSASS memory dumping and Kerberoasting attacks, enabling attackers to obtain and crack domain credentials.
- Privilege escalation was achieved through process hollowing and token impersonation, granting high-level SYSTEM access.
- Attackers conducted extensive reconnaissance and internal discovery, including vulnerability scanning, process enumeration, and Active Directory exploration.
- Defense evasion techniques included disabling security controls, deleting logs, and removing payloads to avoid detection.
- Lateral movement was performed using RDP, while tools like AnyDesk were deployed for persistent command-and-control (C2) access.
- Sensitive data was collected, archived, and exfiltrated using covert techniques such as DNS tunneling and tools like dnscat2.
- The attack targeted critical infrastructure including Active Directory, indicating intent for full domain compromise.
- Overall risk level is MEDIUM-HIGH (74%), with high potential impact due to credential theft, data exfiltration, and persistent attacker presence.
- Immediate response actions such as system isolation, credential resets, and forensic investigation are strongly recommended to prevent further damage.

II

CASE IDENTIFICATION & BACKGROUND

Attack Narrative: Registry Run Key Persistence + LSASS Credential Dump ; Active Vulnerability Scanning and Reconnaissance ; Web Shell Exploit to Scheduled Task Persistence ; Process Hollowing and Token Impersonation

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-DC-01, WIN-SOC-PROD-07, WIN-SOC-PROD-08
Users	WIN-SOC\User1, WIN-SOC\Dev-01, NT AUTHORITY\SYSTEM, WIN-SOC\Admin, WIN-SOC\WebSvc, WIN-SOC\SvcOps, WIN-SOC\Analyst2, WIN-SOC\DBAdmin, WIN-SOC\ITSupport

III DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

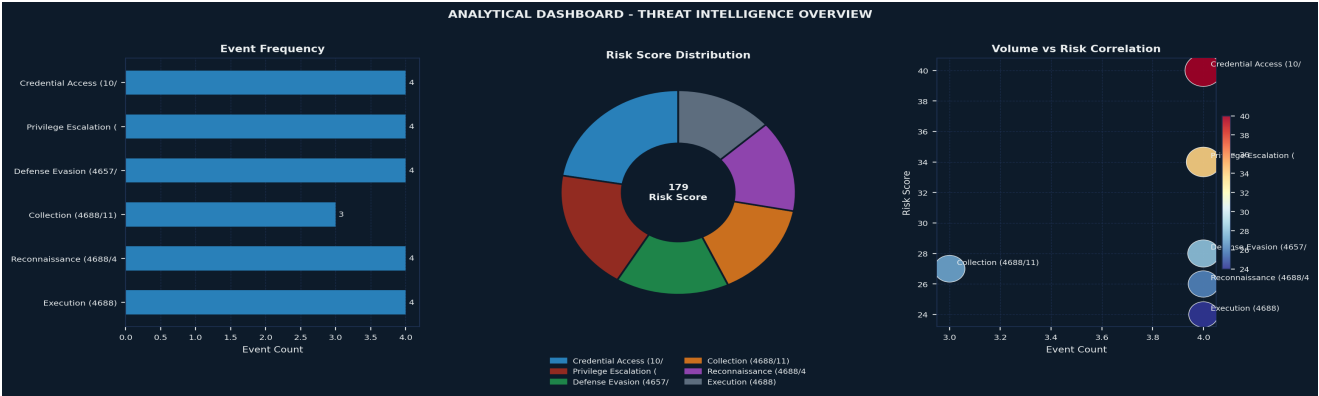


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

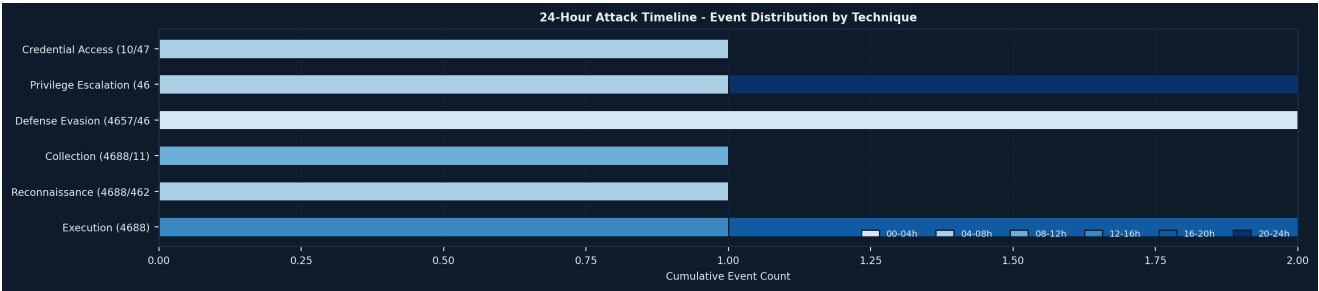


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (10/4769)

LOW

Events: 4	Risk Index: 40	MITRE: T1003.001, T1558.003 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-03-27 23:42:35	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-03-27 23:42:49	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-03-28 04:02:30	WIN-SOC-DC-01	WIN-SOC\DBAdmin	4769
2026-03-28 04:03:10	WIN-SOC-DC-01	WIN-SOC\DBAdmin	4769

Analysis: Credential Access (10/4769) produced 4 suspicious artifacts with cumulative risk index 40. The mapped technique T1003.001, T1558.003 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4657/4688)

LOW

Events: 4	Risk Index: 28	MITRE: T1112, T1070.004 Tactic: Defense Evasion	
Timestamp	Host	User Context	EID
2026-03-28 03:00:00	WIN-SOC-PROD-05	WIN-SOC\Analyst2	4657
2026-03-28 03:00:40	WIN-SOC-PROD-05	WIN-SOC\Analyst2	4657
2026-03-28 03:01:20	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688
2026-03-28 03:02:00	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Defense Evasion (4657/4688) produced 4 suspicious artifacts with cumulative risk index 28. The mapped technique T1112, T1070.004 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4624/4688)

LOW

Events: 3	Risk Index: 23	MITRE: T1021.001 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-03-28 05:00:00	WIN-SOC-PROD-07	WIN-SOC\ITSupport	4624
2026-03-28 05:00:40	WIN-SOC-PROD-07	WIN-SOC\ITSupport	4688
2026-03-28 05:03:00	WIN-SOC-PROD-08	WIN-SOC\ITSupport	4688

Analysis: Lateral Movement (4624/4688) produced 3 suspicious artifacts with cumulative risk index 23. The mapped technique T1021.001 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688/8)

LOW

Events: 4	Risk Index: 34	MITRE: T1055.002, T1134.001 Tactic: Privilege Escalation	
Timestamp	Host	User Context	EID
2026-03-28 02:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688
2026-03-28 02:05:30	WIN-SOC-PROD-04	WIN-SOC\SvcOps	8
2026-03-28 02:06:10	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-03-28 02:07:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4672

Analysis: Privilege Escalation (4688/8) produced 4 suspicious artifacts with cumulative risk index 34. The mapped technique T1055.002, T1134.001 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (13/4698)

LOW

Events: 3	Risk Index: 19	MITRE: T1547.001, T1053.005 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-03-27 23:41:48	WIN-SOC-PROD-01	WIN-SOC\Dev-01	13
2026-03-27 23:41:49	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	13
2026-03-28 01:02:00	WIN-SOC-PROD-03	WIN-SOC\WebSvc	4698

Analysis: Persistence (13/4698) produced 3 suspicious artifacts with cumulative risk index 19. The mapped technique T1547.001, T1053.005 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-01	9	2	LOW
WIN-SOC-PROD-03	6	4	MEDIUM

WIN-SOC-PROD-08	6	6	MEDIUM
WIN-SOC-PROD-04	4	3	LOW
WIN-SOC-PROD-05	4	2	LOW
WIN-SOC-PROD-07	4	3	LOW
WIN-SOC-PROD-06	3	1	LOW
WIN-SOC-DC-01	3	2	LOW
WIN-SOC-PROD-02	1	0	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Registry Run Key Persistence + LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

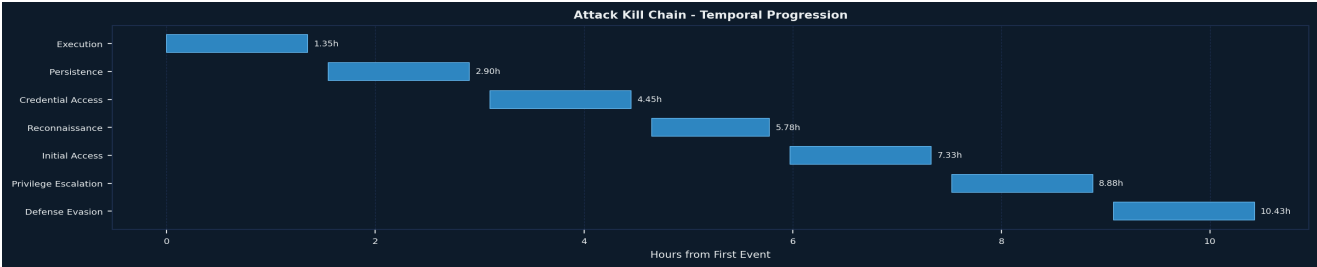
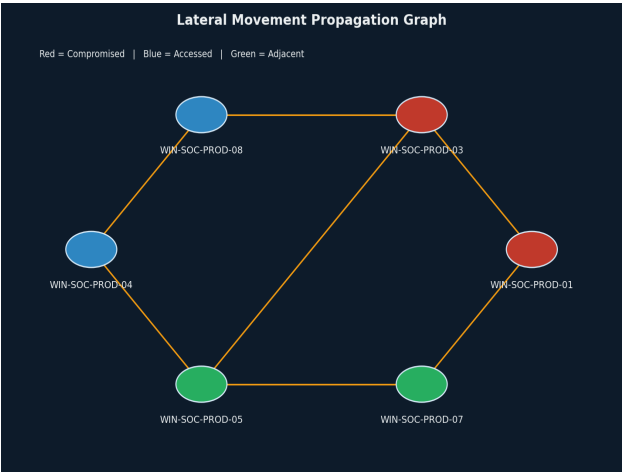


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-005-01	Encoded IEX PowerShell Download → Registry Run Key Written (HKCU + HKLM) → LSASS Memory Dumped (mimikatz.exe)	T1059.001, T1547.001, T1003.001
chain-ds-005-02	nmap Vulnerability Scan → masscan Mass Port Scan → nikto Web App Scan → Logon Failures on Target Host	T1595.002, T1595.002, T1595.002
chain-ds-005-03	IIS w3wp.exe Web Shell → cmd.exe Reverse Shell → PowerShell Dropper → Scheduled Task Created for Persistence → Attacker Logon Confirmed	T1190, T1059.001, T1053.005

chain-ds-0 05-04	svchost.exe Process Hollowing → Remote Thread Injection → whoami /priv Confirmation → SelImpersonatePrivilege Token Theft	T1055.002, T1134.001, T1055.002
chain-ds-0 05-05	Firewall Disabled via Registry → UAC Disabled via Registry → Security Logs Deleted → Payload Files Removed After Execution	T1112, T1112, T1070.004
chain-ds-0 05-06	tasklist + Get-Process Enumeration → Config File Discovery → AD dsquery Enumeration → Kerberoasting SPN Ticket Requests → RC4 Encryption Downgrade	T1057, T1057, T1558.003
chain-ds-0 05-07	RDP Logon from Non-Standard Host → RDP Session PROD-06 to PROD-07 → AnyDesk Installed for Persistent C2 → AnyDesk Outbound Beacon → RDP Pivot to PROD-08	T1021.001, T1021.001, T1219
chain-ds-0 05-08	Snipping Tool Automated Screenshot Collection → Screenshot Files Written to Disk → Data Archived with 7zip → DNS TXT Query Exfiltration → dnscat2 C2 Tunnel Exfiltration	T1113, T1560.001, T1048

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	reg.exe	Registry Run Key Written: HKCU\Software\Microsoft\Windows\CurrentVersion\Run\UpdateHelper
Event ID	13	Sysmon
Process	mimikatz.exe	LSASS Dump
Event ID	10	Sysmon
Process	nmap.exe	Vulnerability Scan Against External Subnet 192.168.10.0/24
Process	masscan.exe	Mass Port Scan Targeting Internet-Facing Services
Process	nikto.exe	Web Application Vulnerability Scan on Port 443
Process	lsass.exe	Repeated Logon Failures from Scanning Source IP 192.168.10.55
Event ID	4625	Security
Process	w3wp.exe	IIS Worker Process Spawned cmd.exe - Web Shell Execution

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

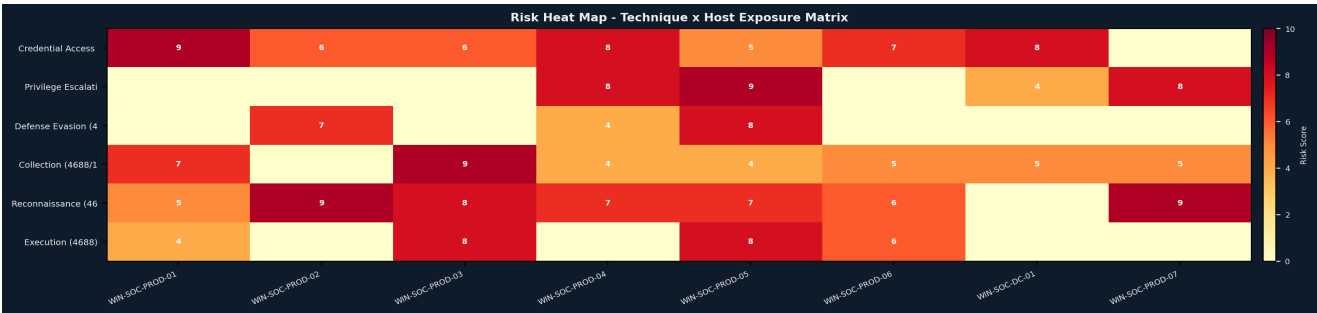


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

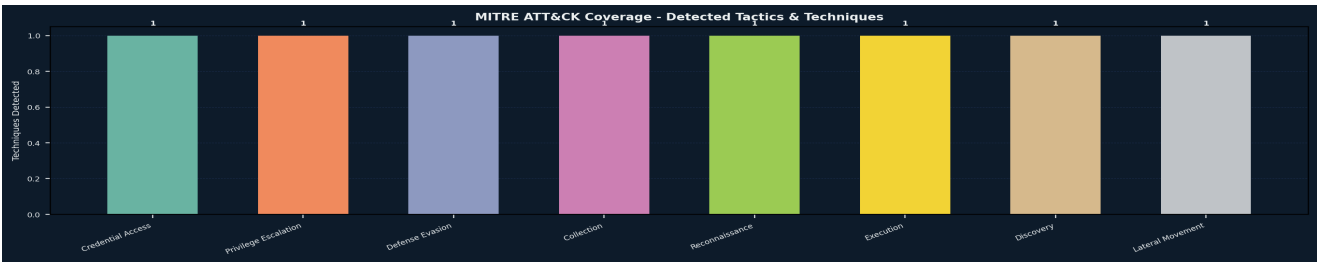


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001, T1558.003	Credential Access (10/4769)	Credential Access
T1055.002, T1134.001	Privilege Escalation (4688/8)	Privilege Escalation
T1112, T1070.004	Defense Evasion (4657/4688)	Defense Evasion
T1113, T1560.001	Collection (4688/11)	Collection
T1595.002	Reconnaissance (4688/4625)	Reconnaissance
T1059.001	Execution (4688)	Execution
T1057	Discovery (4688/4663)	Discovery
T1021.001	Lateral Movement (4624/4688)	Lateral Movement

XVII

REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

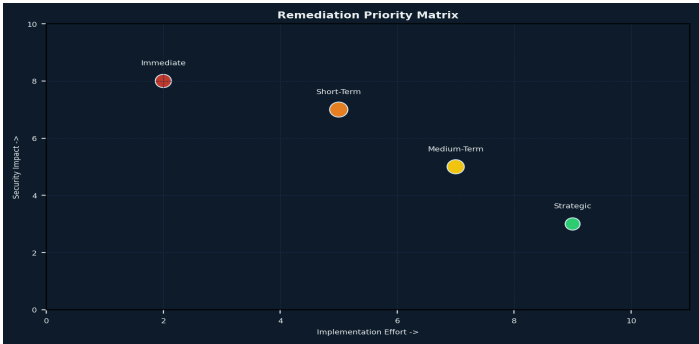


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII

EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset5_expanded**, the environment exhibits a **HIGH** severity posture with aggregate risk index **7,400** across **10,000** records. Dominant indicators include **Credential Access (10/4769) (4 events)**, **Privilege Escalation (4688/8) (4 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst

Automated Forensic Engine

Authorizing Officer

Incident Response Division

XIX

ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:57:31

XX

ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-03-27 23:41:45	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-27 23:41:46	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-27 23:41:47	WIN-SOC-PROD-02	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-27 23:41:48	WIN-SOC-PROD-01	WIN-SOC\Dev-01	reg.exe	Registry Run Key Written: HKCU\Software\Microsoft\Windows\CurrentVersion\Run\UpdateHelper
2026-03-27 23:41:49	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	reg.exe	Registry Run Key Written: HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SysMon32
2026-03-27 23:42:35	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-27 23:42:49	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-28 00:15:00	WIN-SOC-PROD-01	WIN-SOC\Admin	nmap.exe	Vulnerability Scan Against External Subnet 192.168.10.0/24
2026-03-28 00:15:40	WIN-SOC-PROD-01	WIN-SOC\Admin	masscan.exe	Mass Port Scan Targeting Internet-Facing Services
2026-03-28 00:16:20	WIN-SOC-PROD-01	WIN-SOC\Admin	nikto.exe	Web Application Vulnerability Scan on Port 443
2026-03-28 00:17:10	WIN-SOC-PROD-03	WIN-SOC\WebSvc	lsass.exe	Repeated Logon Failures from Scanning Source IP 192.168.10.55
2026-03-28 01:00:10	WIN-SOC-PROD-03	WIN-SOC\WebSvc	w3wp.exe	IIS Worker Process Spawned cmd.exe - Web Shell Execution
2026-03-28 01:00:45	WIN-SOC-PROD-03	WIN-SOC\WebSvc	cmd.exe	Web Shell Reverse Shell via cmd.exe
2026-03-28 01:01:20	WIN-SOC-PROD-03	WIN-SOC\WebSvc	powershell.exe	Post-Exploitation PowerShell Dropper via Web Shell
2026-03-28 01:02:00	WIN-SOC-PROD-03	WIN-SOC\WebSvc	schtasks.exe	Scheduled Task Created for Web Shell Persistence: \IISAppPoolMaintenance
2026-03-28 01:03:00	WIN-SOC-PROD-03	WIN-SOC\WebSvc	lsass.exe	Attacker Logon After Web Shell Access
2026-03-28 02:05:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	svchost.exe	Process Hollowing: svchost.exe Replaced with Malicious Payload
2026-03-28 02:05:30	WIN-SOC-PROD-04	WIN-SOC\SvcOps	svchost.exe	Remote Thread Injected into svchost.exe via Process Hollowing
2026-03-28 02:06:10	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	whoami.exe	Token Impersonation Confirmed via whoami /priv
2026-03-28 02:07:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	lsass.exe	SeImpersonatePrivilege Assigned - Token Theft Confirmed

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.